

# **HW 2: 1D Gaussian Filtering with Wrap Condition**

과목 : 기초 GPU 프로그래밍

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## 실험 환경

GPU : Geforce RTX 4070 Ti

CUDA Compute Capability : 8.9

## 측정 방법

커널 함수의 수행 시간은 dummy call 이후 32회 반복 수행한 결과의 평균값으로 측정하였다.

### - 세 가지 구현 커널 각각에 대한 register 사용량

Register 사용량은 NSight Compute의 launch\_registers\_per\_thread 값을 통해 측정하였다. 해당 값은 Compile time에 결정되므로 Thread Block 크기와 무관하게 동일한 값을 가진다.

### - Thread Block 크기에 따른 Shared Memory 사용량

Shared Memory 사용량은 launch\_shared\_mem\_per\_block 값을 통해 확인하였다. ORIGINAL 및 STRIDE 커널은 shared memory를 사용하지 않으며 SHARED 커널의 경우 Thread Block 크기와 kernel\_width에 따라 dynamic shared memory 크기가 변화하는 것을 비교하였다.

### - Thread block 크기에 따른 achieved (warp) occupancy 변화

Achieved (Warp) Occupancy는 sm\_maximum\_warps\_per\_active\_cycle\_pct 값을 기준으로 측정하였다. 이 값은 SM이 이론적으로 처리 가능한 최대 warp 수 대비 실제 활성 warp 수의 비율을 나타내며 Thread Block 크기별로 변화 추이를 비교하였다.

### - 각종 throughput 포함 기타 프로파일링 항목

SM Throughput 및 기타 항목은 sm\_throughput.avg.pct\_of\_peak\_sustained\_elapsed 값을 사용하여 SM의 이론적 최대 처리량 대비 실제 활용률을 비교했다. 추가적으로 L1 캐시 및 DRAM 접근 관련 throughput 값을 통해 각 커널의 특성을 파악하였다.

측정 결과 및 분석

1D Gaussian Filtering 성능 측정 결과 (32회 평균)

Threads per Block = 64

| Kernel Width | ORIGINAL  | SHARED   | STRIDE(2) | STRIDE(4) | STRIDE(8) | STRIDE(16) | STRIDE(32) |
|--------------|-----------|----------|-----------|-----------|-----------|------------|------------|
| 1            | 1.533 ms  | 1.806 ms | 1.414 ms  | 1.299 ms  | 1.309 ms  | 1.334 ms   | 1.374 ms   |
| 3            | 2.453 ms  | 3.132 ms | 2.022 ms  | 1.883 ms  | 1.821 ms  | 1.795 ms   | 1.818 ms   |
| 5            | 3.286 ms  | 3.917 ms | 2.769 ms  | 2.629 ms  | 2.566 ms  | 2.547 ms   | 2.560 ms   |
| 7            | 3.948 ms  | 4.467 ms | 3.612 ms  | 3.493 ms  | 3.409 ms  | 3.393 ms   | 3.411 ms   |
| 16           | 5.084 ms  | 5.214 ms | 4.904 ms  | 4.616 ms  | 4.477 ms  | 4.248 ms   | 4.297 ms   |
| 11           | 5.767 ms  | 5.954 ms | 5.347 ms  | 5.232 ms  | 5.155 ms  | 5.152 ms   | 5.232 ms   |
| 13           | 10.423 ms | 6.713 ms | 6.199 ms  | 6.051 ms  | 6.012 ms  | 6.007 ms   | 6.038 ms   |
| 33           | 10.980 ms | 7.454 ms | 7.039 ms  | 6.897 ms  | 6.834 ms  | 6.891 ms   | 6.929 ms   |

Threads per Block = 128

| Kernel Width | ORIGINAL | SHARED   | STRIDE(2) | STRIDE(4) | STRIDE(8) | STRIDE(16) | STRIDE(32) |
|--------------|----------|----------|-----------|-----------|-----------|------------|------------|
| 1            | 1.660 ms | 1.866 ms | 1.320 ms  | 1.297 ms  | 1.329 ms  | 1.367 ms   | 1.375 ms   |
| 3            | 2.447 ms | 3.137 ms | 2.048 ms  | 1.961 ms  | 1.840 ms  | 1.847 ms   | 1.841 ms   |
| 5            | 3.190 ms | 3.708 ms | 2.912 ms  | 2.730 ms  | 2.671 ms  | 2.641 ms   | 2.700 ms   |
| 7            | 4.454 ms | 4.623 ms | 3.793 ms  | 3.654 ms  | 3.540 ms  | 3.588 ms   | 3.883 ms   |
| 25           | 5.426 ms | 5.139 ms | 4.712 ms  | 4.551 ms  | 4.314 ms  | 4.366 ms   | 4.341 ms   |
| 11           | 5.973 ms | 6.028 ms | 5.654 ms  | 5.715 ms  | 5.483 ms  | 5.166 ms   | 5.163 ms   |
| 13           | 6.647 ms | 6.431 ms | 6.221 ms  | 6.156 ms  | 6.013 ms  | 6.155 ms   | 6.263 ms   |
| 15           | 8.585 ms | 7.818 ms | 7.090 ms  | 6.947 ms  | 6.856 ms  | 7.003 ms   | 7.275 ms   |

## Threads per Block = 256

| Kernel Width | ORIGINAL | SHARED   | STRIDE(2) | STRIDE(4) | STRIDE(8) | STRIDE(16) | STRIDE(32) |
|--------------|----------|----------|-----------|-----------|-----------|------------|------------|
| 1            | 1.507 ms | 1.792 ms | 1.592 ms  | 1.269 ms  | 1.320 ms  | 1.323 ms   | 1.357 ms   |
| 3            | 2.298 ms | 3.028 ms | 2.109 ms  | 1.913 ms  | 1.846 ms  | 1.807 ms   | 1.834 ms   |
| 5            | 3.201 ms | 3.767 ms | 2.909 ms  | 2.745 ms  | 2.667 ms  | 2.647 ms   | 2.648 ms   |
| 7            | 4.188 ms | 4.578 ms | 3.811 ms  | 3.667 ms  | 3.656 ms  | 3.578 ms   | 3.569 ms   |
| 9            | 4.989 ms | 5.265 ms | 4.648 ms  | 4.352 ms  | 4.448 ms  | 4.376 ms   | 4.432 ms   |
| 11           | 5.868 ms | 6.137 ms | 5.572 ms  | 5.846 ms  | 5.384 ms  | 5.564 ms   | 5.443 ms   |
| 13           | 6.840 ms | 7.156 ms | 7.253 ms  | 6.358 ms  | 5.999 ms  | 6.150 ms   | 6.251 ms   |
| 15           | 8.311 ms | 7.479 ms | 7.033 ms  | 6.985 ms  | 7.086 ms  | 6.853 ms   | 6.943 ms   |

## Threads per Block = 512

| Kernel Width | ORIGINAL | SHARED   | STRIDE(2) | STRIDE(4) | STRIDE(8) | STRIDE(16) | STRIDE(32) |
|--------------|----------|----------|-----------|-----------|-----------|------------|------------|
| 1            | 1.739 ms | 1.870 ms | 1.426 ms  | 1.280 ms  | 1.305 ms  | 1.270 ms   | 1.288 ms   |
| 3            | 2.344 ms | 3.023 ms | 2.075 ms  | 1.938 ms  | 1.821 ms  | 1.769 ms   | 1.788 ms   |
| 5            | 3.266 ms | 3.811 ms | 2.918 ms  | 2.679 ms  | 2.610 ms  | 2.579 ms   | 2.635 ms   |
| 7            | 4.225 ms | 4.542 ms | 3.762 ms  | 3.665 ms  | 3.534 ms  | 3.513 ms   | 3.853 ms   |
| 9            | 4.974 ms | 5.403 ms | 4.799 ms  | 5.384 ms  | 4.836 ms  | 4.814 ms   | 4.310 ms   |
| 11           | 6.143 ms | 6.283 ms | 5.481 ms  | 5.236 ms  | 5.465 ms  | 5.630 ms   | 6.037 ms   |
| 13           | 6.875 ms | 6.952 ms | 6.390 ms  | 6.078 ms  | 6.383 ms  | 6.613 ms   | 6.219 ms   |
| 15           | 7.701 ms | 7.469 ms | 7.135 ms  | 6.969 ms  | 6.938 ms  | 7.574 ms   | 7.241 ms   |

## Threads per Block = 1024

| Kernel Width | ORIGINAL  | SHARED    | STRIDE(2) | STRIDE(4) | STRIDE(8) | STRIDE(16) | STRIDE(32) |
|--------------|-----------|-----------|-----------|-----------|-----------|------------|------------|
| 1            | 2.439 ms  | 2.717 ms  | 1.772 ms  | 1.620 ms  | 1.541 ms  | 1.490 ms   | 1.397 ms   |
| 3            | 3.243 ms  | 4.425 ms  | 2.793 ms  | 2.594 ms  | 2.469 ms  | 2.407 ms   | 2.480 ms   |
| 5            | 4.400 ms  | 5.405 ms  | 3.907 ms  | 3.647 ms  | 3.525 ms  | 3.467 ms   | 3.498 ms   |
| 7            | 5.550 ms  | 6.360 ms  | 4.977 ms  | 4.752 ms  | 4.617 ms  | 4.614 ms   | 4.657 ms   |
| 9            | 6.624 ms  | 7.341 ms  | 6.078 ms  | 5.824 ms  | 5.711 ms  | 5.742 ms   | 5.698 ms   |
| 11           | 7.757 ms  | 8.311 ms  | 7.253 ms  | 6.959 ms  | 6.828 ms  | 6.778 ms   | 6.860 ms   |
| 13           | 9.211 ms  | 9.803 ms  | 8.834 ms  | 8.283 ms  | 8.512 ms  | 8.827 ms   | 7.998 ms   |
| 15           | 10.289 ms | 10.601 ms | 10.629 ms | 9.537 ms  | 9.224 ms  | 9.268 ms   | 9.061 ms   |

## 세 가지 구현 커널 각각에 대한 Register 사용량

NSight Compute에서 `launch_registers_per_thread`으로 확인한 각 커널의 레지스터 사용량을 분석했다. 레지스터 수는 thread block 크기와 무관하게 동일하게 유지된다. 레지스터 할당은 Compile Time에 결정되며 Runtime Parameter에 의존하지 않기 때문이다.

| 커널       | Pred Reg | F32 Reg | B32 Reg | B64 Reg |
|----------|----------|---------|---------|---------|
| ORIGINAL | 16       | 27      | 53      | 23      |
| SHARED   | 10       | 33      | 71      | 23      |
| STRIDE   | 25       | 31      | 102     | 39      |

ORIGINAL 커널은 Global Memory에서 직접 데이터를 읽어 연산하므로 인덱스 계산에 필요한 Register만 사용한다. SHARED 커널은 halo 영역을 포함한 Shared Memory 로딩 로직(경계 조건 처리, 좌우 halo 각각 별도 thread 처리)으로 인해 predicate 레지스터와 B32 Register가 증가한다. STRIDE 커널은 grid-stride loop를 위한 stride 계산 변수가 추가되어 ORIGINAL보다 약간 많은 Register를 사용한다.

## Thread Block 크기에 따른 Shared Memory 사용량

ORIGINAL 및 STRIDE 커널은 shared memory를 사용하지 않는다. SHARED 커널은 dynamic shared memory를 사용하며 그 크기는  $(TPB + 2 \times \text{kernel\_radius}) \times 4 \text{ bytes}$ 로 계산된다. (Kernel radius는 kernel width의 반이다.  $\text{kernel radius} = (\text{kernel\_width} - 1) / 2$ )

이 표는 대표적인 kernel width 값에 대한 shared memory 크기를 나타낸다.

| Thread<br>per<br>block | Kernel<br>Width<br>=1 | Kernel<br>Width<br>=5 | Kernel<br>Width<br>=9 | Kernel<br>Width<br>=15 | Shared Memory<br>사용량 범위 |
|------------------------|-----------------------|-----------------------|-----------------------|------------------------|-------------------------|
| 64                     | 256 B                 | 272 B                 | 288 B                 | 312 B                  | 256 ~ 312 B             |
| 128                    | 512 B                 | 528 B                 | 544 B                 | 568 B                  | 512 ~ 568 B             |
| 256                    | 1024 B                | 1040 B                | 1056 B                | 1080 B                 | 1024 ~ 1080 B           |
| 512                    | 2048 B                | 2064 B                | 2080 B                | 2104 B                 | 2048 ~ 2104 B           |
| 1024                   | 4096 B                | 4112 B                | 4128 B                | 4152 B                 | 4096 ~ 4152 B           |

Original, Stride Kernel은 Global Memory에 직접 접근하기에 shared memory 사용량이 0이다.

Shared memory 사용량은 TPB에 비례하여 증가하며 kernel width가 커질수록 halo 크기( $2 \times \text{radius}$ )만큼 추가된다. TPB=1024, KW=15의 경우 4216 bytes를 사용한다. CC 8.9의 SM당 최대 shared memory는 100 KB이므로 단일 커널 기준으로는 shared memory가 occupancy 병목이 되지 않는다. 그러나 TPB=1024에서는 thread block 수가 줄어들어 SM 활용률이 함께 감소한다.

## Thread Block 크기에 따른 Achieved (Warp) Occupancy 변화

CC 8.9 기준 SM당 최대 warp 수는 48이다. NSight Compute의 `sm_maximum_warps_per_active_cycle_pct` 및 `smssp_maximum_warps_avg_per_active_cycle`으로 측정한다.

| TPB | ORIGINAL (%) | SHARED (%) | STRIDE (%) | 제한 요인                                 |
|-----|--------------|------------|------------|---------------------------------------|
| 64  | ~100%        | ~100%      | ~100%      | Block 수 충분,<br>register 제한 없음 → ~100% |
| 128 | ~100%        | ~100%      | ~100%      | Block 수 충분 → ~100%                    |

| TPB  | ORIGINAL (%) | SHARED (%) | STRIDE (%) | 제한 요인                                                                         |
|------|--------------|------------|------------|-------------------------------------------------------------------------------|
| 256  | ~100%        | ~100%      | ~100%      | Thread 수 제한 ( $1536/256=6$ blocks)<br>$\times 8$ warps = $48/48 = 100\%$      |
| 512  | ~100%        | ~100%      | ~100%      | Thread 수 제한 ( $1536/512=3$ blocks)<br>$\times 16$ warps = $48/48 = 100\%$     |
| 1024 | ~67%         | ~67%       | ~67%       | Thread 수 제한 ( $1536/1024=1$ block)<br>$\times 32$ warps = $32/48 = \sim 67\%$ |

CC 8.9 기준 SM당 최대 1536 threads 제한으로 인해 TPB 64~512에서는 충분한 수의 block이 SM에 배치되어 48 warps를 채울 수 있으므로 occupancy가 ~100%로 유지된다. TPB=1024에서는 SM당 배치 가능한 block 수가 1개로 제한되어 32 warps만 활성화되므로 occupancy가  $32/48 \approx 67\%$ 로 감소한다. 이는 TPB=1024에서의 성능 저하와 직접 연관된다. SHARED 커널은 register 수가 더 많지만 CC 8.9의 register 파일(65536개/SM) 때문에 TPB 64~512 범위에서 occupancy 병목이 발생하지 않는다.

## SM Throughput 및 기타 프로파일링 항목

sm\_throughput.avg.pct\_of\_peak\_sustained\_elapsed은 SM이 이론적 최대 처리량 대비 실제로 처리한 비율을 나타낸다.

| TPB  | ORIGINAL (%) | SHARED (%) | STRIDE (%) | 특징                                                        |
|------|--------------|------------|------------|-----------------------------------------------------------|
| 64   | 78.8%        | 79.6%      | 78.5%      | 세 커널 모두 높은 SM 활용률                                         |
| 128  | 79.2%        | 79.6%      | 78.7%      | TPB=64와 유사한 패턴                                            |
| 256  | 79.2%        | 78.5%      | 77.5%      | SHARED SM throughput 소폭 감소                                |
| 512  | 77.4%        | 76.8%      | 76.9%      | 전반적 throughput 저하, occupancy 감소 영향                        |
| 1024 | 75.0%        | 74.3%      | 75.4%      | SM throughput 가장 낮음.<br>∴ block 수 감소, latency hiding 어려움. |

## - Memory Throughput 분석

1D Gaussian filtering은 각 출력 픽셀마다  $2 \times \text{kernel\_radius} + 1$ 개의 입력 픽셀을 읽는다. Kernel\_width가 커질수록 연산량 대비 memory access 비율이 높아진다. ORIGINAL 커널은 인접 픽셀을 반복적으로 global memory에서 읽어 L2 cache 재사용률은 일부 있으나 이웃 스레드 간 중복 로드가 발생한다. SHARED 커널은 이를 shared memory로 캐싱하여 global memory 접근 횟수를 줄이지만 shared memory 관리 overhead가 추가된다.

## - L1/L2 Cache 활용

ORIGINAL 및 STRIDE 커널에서는 인접 스레드들이 연속된 메모리 주소를 접근하므로 coalesced memory access 패턴을 보인다. L1 cache(l1tex\_throughput)는 연산 중 중간 데이터의 임시 저장에 활용되며 L2 cache(l2s\_throughput)에서의 hit rate는 kernel width와 반비례 관계를 보인다. kernel width가 커질수록 working set이 커져 L2 hit rate가 감소하고 DRAM bandwidth 소모가 증가한다.

## - STRIDE 커널 특이 사항

Grid-stride loop 방식은 하나의 thread가 여러 출력 픽셀을 담당하여 thread block 수를 줄이는 대신 루프 반복으로 작업을 처리하기에 TPB 64에서 STRIDE(4) 기준으로 block 수가 1/4로 줄어들지만 각 thread의 연산량이 증가하여 latency hiding이 향상된다. 그래서 STRIDE 커널이 소규모 stride(2~4)에서 ORIGINAL 대비 동등하거나 낮은 실행 시간을 달성한다.



아래는 프로그램 실행 결과들이다.

```
=== 1D Gaussian filtering of image of 67108867 pixels(KERNEL WIDTH = 1) ===

[HOST] Time to filter an image of 67108867 pixels on the host = 177.270(ms)

[ORIGINAL] CUDA kernel launch with 524289 blocks of 128 threads
Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.660(ms) / speedup = 106.793
Test PASSED

[SHARED] CUDA kernel launch with 524289 blocks of 128 threads
Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.866(ms) / speedup = 94.997
Test PASSED

[STRIDE(2)] CUDA kernel launch with 262144(524289) blocks of 128 threads
Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.320(ms) / speedup = 134.278
Test PASSED

[STRIDE(4)] CUDA kernel launch with 131072(524289) blocks of 128 threads
Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.297(ms) / speedup = 136.699
Test PASSED

[STRIDE(8)] CUDA kernel launch with 65536(524289) blocks of 128 threads
Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.329(ms) / speedup = 133.346
Test PASSED

[STRIDE(16)] CUDA kernel launch with 32768(524289) blocks of 128 threads
Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.367(ms) / speedup = 129.643
Test PASSED

[STRIDE(32)] CUDA kernel launch with 16384(524289) blocks of 128 threads
Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.375(ms) / speedup = 128.878
Test PASSED

=== 1D Gaussian filtering of image of 67108867 pixels(KERNEL WIDTH = 3) ===

[HOST] Time to filter an image of 67108867 pixels on the host = 200.629(ms)

[ORIGINAL] CUDA kernel launch with 524289 blocks of 128 threads
Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.447(ms) / speedup = 82.005
Test PASSED

[SHARED] CUDA kernel launch with 524289 blocks of 128 threads
Average Time to filter an image of 67108867 pixels on the device (32 tests) = 3.137(ms) / speedup = 63.957
Test PASSED

[STRIDE(2)] CUDA kernel launch with 262144(524289) blocks of 128 threads
Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.048(ms) / speedup = 97.959
Test PASSED

[STRIDE(4)] CUDA kernel launch with 131072(524289) blocks of 128 threads
Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.961(ms) / speedup = 102.310
Test PASSED

[STRIDE(8)] CUDA kernel launch with 65536(524289) blocks of 128 threads
Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.840(ms) / speedup = 109.056
Test PASSED

[STRIDE(16)] CUDA kernel launch with 32768(524289) blocks of 128 threads
Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.847(ms) / speedup = 108.645
Test PASSED

[STRIDE(32)] CUDA kernel launch with 16384(524289) blocks of 128 threads
Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.841(ms) / speedup = 108.973
Test PASSED

=== 1D Gaussian filtering of image of 67108867 pixels(KERNEL WIDTH = 5) ===
```

=== 1D Gaussian filtering of image of 67108867 pixels(KERNEL WIDTH = 1) ===

[HOST] Time to filter an image of 67108867 pixels on the host = 192.361(ms)

[ORIGINAL] CUDA kernel launch with 1048577 blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.533(ms) / speedup = 125.512  
Test PASSED

[SHARED] CUDA kernel launch with 1048577 blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.806(ms) / speedup = 106.517  
Test PASSED

[STRIDE(2)] CUDA kernel launch with 524288(1048577) blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.414(ms) / speedup = 136.002  
Test PASSED

[STRIDE(4)] CUDA kernel launch with 262144(1048577) blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.299(ms) / speedup = 148.085  
Test PASSED

[STRIDE(8)] CUDA kernel launch with 131072(1048577) blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.309(ms) / speedup = 146.928  
Test PASSED

[STRIDE(16)] CUDA kernel launch with 65536(1048577) blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.334(ms) / speedup = 144.228  
Test PASSED

[STRIDE(32)] CUDA kernel launch with 32768(1048577) blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.374(ms) / speedup = 139.970  
Test PASSED

=== 1D Gaussian filtering of image of 67108867 pixels(KERNEL WIDTH = 3) ===

[HOST] Time to filter an image of 67108867 pixels on the host = 444.584(ms)

[ORIGINAL] CUDA kernel launch with 1048577 blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.453(ms) / speedup = 181.263  
Test PASSED

[SHARED] CUDA kernel launch with 1048577 blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 3.132(ms) / speedup = 141.939  
Test PASSED

[STRIDE(2)] CUDA kernel launch with 524288(1048577) blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.022(ms) / speedup = 219.916  
Test PASSED

[STRIDE(4)] CUDA kernel launch with 262144(1048577) blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.883(ms) / speedup = 236.134  
Test PASSED

[STRIDE(8)] CUDA kernel launch with 131072(1048577) blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.821(ms) / speedup = 244.146  
Test PASSED

[STRIDE(16)] CUDA kernel launch with 65536(1048577) blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.795(ms) / speedup = 247.735  
Test PASSED

[STRIDE(32)] CUDA kernel launch with 32768(1048577) blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.818(ms) / speedup = 244.604  
Test PASSED

=== 1D Gaussian filtering of image of 67108867 pixels(KERNEL WIDTH = 5) ===

[HOST] Time to filter an image of 67108867 pixels on the host = 730.326(ms)

[ORIGINAL] CUDA kernel launch with 1048577 blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 3.286(ms) / speedup = 222.235  
Test PASSED

[SHARED] CUDA kernel launch with 1048577 blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 3.917(ms) / speedup = 186.438  
Test PASSED

[STRIDE(2)] CUDA kernel launch with 524288(1048577) blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.769(ms) / speedup = 263.724  
Test PASSED

[STRIDE(4)] CUDA kernel launch with 262144(1048577) blocks of 64 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.629(ms) / speedup = 277.833  
Test PASSED

=== 1D Gaussian filtering of image of 67108867 pixels(KERNEL WIDTH = 1) ===

[HOST] Time to filter an image of 67108867 pixels on the host = 129.042(ms)

[ORIGINAL] CUDA kernel launch with 262145 blocks of 256 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.507(ms) / speedup = 85.655  
Test PASSED

[SHARED] CUDA kernel launch with 262145 blocks of 256 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.792(ms) / speedup = 72.006  
Test PASSED

[STRIDE(2)] CUDA kernel launch with 131072(262145) blocks of 256 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.592(ms) / speedup = 81.071  
Test PASSED

[STRIDE(4)] CUDA kernel launch with 65536(262145) blocks of 256 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.269(ms) / speedup = 101.665  
Test PASSED

[STRIDE(8)] CUDA kernel launch with 32768(262145) blocks of 256 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.320(ms) / speedup = 97.769  
Test PASSED

[STRIDE(16)] CUDA kernel launch with 16384(262145) blocks of 256 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.323(ms) / speedup = 97.558  
Test PASSED

[STRIDE(32)] CUDA kernel launch with 8192(262145) blocks of 256 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.357(ms) / speedup = 95.073  
Test PASSED

=== 1D Gaussian filtering of image of 67108867 pixels(KERNEL WIDTH = 3) ===

[HOST] Time to filter an image of 67108867 pixels on the host = 200.982(ms)

[ORIGINAL] CUDA kernel launch with 262145 blocks of 256 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.298(ms) / speedup = 87.443  
Test PASSED

[SHARED] CUDA kernel launch with 262145 blocks of 256 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 3.028(ms) / speedup = 66.379  
Test PASSED

[STRIDE(2)] CUDA kernel launch with 131072(262145) blocks of 256 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.109(ms) / speedup = 95.287  
Test PASSED

[STRIDE(4)] CUDA kernel launch with 65536(262145) blocks of 256 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.913(ms) / speedup = 105.064  
Test PASSED

[STRIDE(8)] CUDA kernel launch with 32768(262145) blocks of 256 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.846(ms) / speedup = 108.861  
Test PASSED

[STRIDE(16)] CUDA kernel launch with 16384(262145) blocks of 256 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.807(ms) / speedup = 111.243  
Test PASSED

[STRIDE(32)] CUDA kernel launch with 8192(262145) blocks of 256 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.834(ms) / speedup = 109.592  
Test PASSED

=== 1D Gaussian filtering of image of 67108867 pixels(KERNEL WIDTH = 1) ===

[HOST] Time to filter an image of 67108867 pixels on the host = 147.176(ms)

[ORIGINAL] CUDA kernel launch with 131073 blocks of 512 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.739(ms) / speedup = 84.646  
Test PASSED

[SHARED] CUDA kernel launch with 131073 blocks of 512 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.870(ms) / speedup = 78.716  
Test PASSED

[STRIDE(2)] CUDA kernel launch with 65536(131073) blocks of 512 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.426(ms) / speedup = 103.192  
Test PASSED

[STRIDE(4)] CUDA kernel launch with 32768(131073) blocks of 512 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.280(ms) / speedup = 114.970  
Test PASSED

[STRIDE(8)] CUDA kernel launch with 16384(131073) blocks of 512 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.305(ms) / speedup = 112.812  
Test PASSED

[STRIDE(16)] CUDA kernel launch with 8192(131073) blocks of 512 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.270(ms) / speedup = 115.858  
Test PASSED

[STRIDE(32)] CUDA kernel launch with 4096(131073) blocks of 512 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.288(ms) / speedup = 114.284  
Test PASSED

=== 1D Gaussian filtering of image of 67108867 pixels(KERNEL WIDTH = 3) ===

[HOST] Time to filter an image of 67108867 pixels on the host = 207.633(ms)

[ORIGINAL] CUDA kernel launch with 131073 blocks of 512 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.344(ms) / speedup = 88.599  
Test PASSED

[SHARED] CUDA kernel launch with 131073 blocks of 512 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 3.023(ms) / speedup = 68.680  
Test PASSED

[STRIDE(2)] CUDA kernel launch with 65536(131073) blocks of 512 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.075(ms) / speedup = 100.087  
Test PASSED

[STRIDE(4)] CUDA kernel launch with 32768(131073) blocks of 512 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.938(ms) / speedup = 107.140  
Test PASSED

[STRIDE(8)] CUDA kernel launch with 16384(131073) blocks of 512 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.821(ms) / speedup = 114.029  
Test PASSED

[STRIDE(16)] CUDA kernel launch with 8192(131073) blocks of 512 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.769(ms) / speedup = 117.363  
Test PASSED

[STRIDE(32)] CUDA kernel launch with 4096(131073) blocks of 512 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.788(ms) / speedup = 116.139  
Test PASSED

=== 1D Gaussian filtering of image of 67108867 pixels(KERNEL WIDTH = 1) ===

[HOST] Time to filter an image of 67108867 pixels on the host = 162.605(ms)

[ORIGINAL] CUDA kernel launch with 65537 blocks of 1024 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.439(ms) / speedup = 66.663  
Test PASSED

[SHARED] CUDA kernel launch with 65537 blocks of 1024 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.717(ms) / speedup = 59.844  
Test PASSED

[STRIDE(2)] CUDA kernel launch with 32768(65537) blocks of 1024 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.772(ms) / speedup = 91.785  
Test PASSED

[STRIDE(4)] CUDA kernel launch with 16384(65537) blocks of 1024 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.620(ms) / speedup = 100.364  
Test PASSED

[STRIDE(8)] CUDA kernel launch with 8192(65537) blocks of 1024 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.541(ms) / speedup = 105.515  
Test PASSED

[STRIDE(16)] CUDA kernel launch with 4096(65537) blocks of 1024 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.490(ms) / speedup = 109.165  
Test PASSED

[STRIDE(32)] CUDA kernel launch with 2048(65537) blocks of 1024 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 1.397(ms) / speedup = 116.354  
Test PASSED

=== 1D Gaussian filtering of image of 67108867 pixels(KERNEL WIDTH = 3) ===

[HOST] Time to filter an image of 67108867 pixels on the host = 202.046(ms)

[ORIGINAL] CUDA kernel launch with 65537 blocks of 1024 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 3.243(ms) / speedup = 62.310  
Test PASSED

[SHARED] CUDA kernel launch with 65537 blocks of 1024 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 4.425(ms) / speedup = 45.656  
Test PASSED

[STRIDE(2)] CUDA kernel launch with 32768(65537) blocks of 1024 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.793(ms) / speedup = 72.345  
Test PASSED

[STRIDE(4)] CUDA kernel launch with 16384(65537) blocks of 1024 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.594(ms) / speedup = 77.880  
Test PASSED

[STRIDE(8)] CUDA kernel launch with 8192(65537) blocks of 1024 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.469(ms) / speedup = 81.846  
Test PASSED

[STRIDE(16)] CUDA kernel launch with 4096(65537) blocks of 1024 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.407(ms) / speedup = 83.929  
Test PASSED

[STRIDE(32)] CUDA kernel launch with 2048(65537) blocks of 1024 threads

Average Time to filter an image of 67108867 pixels on the device (32 tests) = 2.480(ms) / speedup = 81.482  
Test PASSED